Multimodal deep learning for objective skill assessment in robot-assisted vesico-urethral anastomosis.

Authors: Shafiei Somayeh B, Shadpour Saeed, Dakwar Anthony, Xu Zhaomin, Mohler James L

Journal: Journal of robotic surgery | **Year:** 2026 | **DOI:** 10.1007/s11701-026-03290-z

Link: https://doi.org/10.1007/s11701-026-03290-z (open with institutional access)

Abstract

This study aimed to classify robot-assisted surgery (RAS) skill levels (inexperienced, competent, and experienced) during performance of vesico-urethral anastomosis (VUA) using multimodal physiological signals. We trained Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) models on data collected from 23 RAS (RAS trainees and experienced surgeons) performing two VUAs on animal tissue. The dataset included 116-channel electroencephalogram (EEG) and 20 eye-tracking signals recorded during two VUA subtasks: (1) needle grasping, positioning, and entry; and (2) needle driving with wrist rotation and suture pull-out. Skill levels were rated by three raters using the Robotic Anastomosis Competency Evaluation (RACE) tool. Hyperparameters of the models were tuned using grid search with group 4-fold cross-validation on 16 participants and final model performance was evaluated on data from 7 unseen (held-out test) participants, repeated over 10 iterations. Weighted F-scores for classifying skill level using EEG and eye-tracking data were 0.84 for subtask 1 and 0.89 for subtask 2. Using paired t-tests, high-density EEG (116 channels) significantly outperformed low-density EEG (32 channels) for subtask 1 ($p = 0.001$), with no difference for subtask 2 ($p = 0.15$). Adding eye-tracking data significantly improved classification for subtask 2 ($p = 0.001$), but not for subtask 1 ($p = 0.5$). Multimodal deep learning using EEG and eye-tracking data enabled objective classification of surgical skills during VUA. The benefits of high-density EEG and multimodal integration were task-dependent, underscoring the need to tailor assessment tools to the cognitive and sensorimotor demands of specific surgical subtasks.

Note: Publisher PDF could not be downloaded from this network (publisher access restrictions). This record provides the full citation and abstract; obtain the full text via the DOI link above.